



Checking understanding of prime numbers

Task A: Properties of primes

1) List the factors of the following and decide which are **prime** numbers.

a) 27

$$1 \times 27$$

$$3 \times 9$$

b) 37

$$1 \times 37$$

c) 87

$$1 \times 87$$

$$3 \times 29$$

d) 57

$$1 \times 57$$

$$3 \times 19$$

37 is the only **prime** number.

2) Alex writes out the first seven **prime** numbers:

2, 3, 5, 7, 11, 13, 17

He uses these numbers to make the following statements.

Do you agree with them?

Give examples to support your answer.

Most **prime** numbers are odd.

e.g. I agree, all **prime** numbers are odd except 2.

All numbers ending in 3 are **prime**.

e.g. I don't agree, 33 ends in a 3 but is not **prime** because it has two factor pairs, 1×33 and 3×11 .

Numbers ending in 5 can be **prime**.

e.g. I agree, 5 ends in 5 and is **prime**. However, this is the only **prime** ending in 5 as all other numbers ending in 5 are multiples of 5.

Prime numbers don't end in 9.

e.g. I don't agree, 29 is a **prime** number.

7 and 17 are **prime** so 27 will be too.

e.g. I don't agree, 27 is not **prime** because it has two factor pairs, 1×27 and 3×9 .

e.g. **Prime** numbers always have two factors - always true

e.g. **Prime** numbers end in 9 - sometimes true

e.g. Numbers ending in 0 are **prime** - never true



Task B: Identifying primes

- 1) Find your way through the maze. You can only move vertically and horizontally and can only move to a **prime** number.

	27	65	21	19	21	9	49	17	79	11
	12	61	7	12	29	97	5	21	29	2
	8	17	18	49	53	51	67	91	58	14
	45	53	75	10	79	81	83	11	71	57
	11	89	35	43	3	9	22	65	89	14
	1	16	15	61	55	17	37	49	23	17
Start	13	19	7	31	88	15	19	47	1	36
	2	10	51	27	21	19	61	13	2	11

- 2) Use the numbers on the cards to complete the sentences.

Use each card only once.

9	10	11	12	13	14	15	16
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- a) 9 or 16 has an odd number of factors.
- b) 11 or 13 has exactly two factors.
- c) 10, 14 or 15 has exactly four factors.
- d) 10 or 14 has 2 as a factor.
- e) 11 or 13 is **prime**.
- f) 9 or 16 is square.
- g) 10, 14 or 15 has two pairs of factors.
- h) 12 has exactly six factors.
- i) Write a sentence that is true for all of these numbers, true for some numbers and not true for any numbers.

e.g. All of the numbers have two or more factors.
 Some of the numbers are **prime** numbers.
 None of the numbers have more than six factors.

